

Beyond the Clouds: A Data Scientist's Journey from Exploration to Causal Insight in Aircraft Detection

An analysis of the Military Aircraft Detection Dataset by Syed Haider Ali Fatmi and Zain Ahmed Shah, Lahore University of Management Sciences (LUMS).



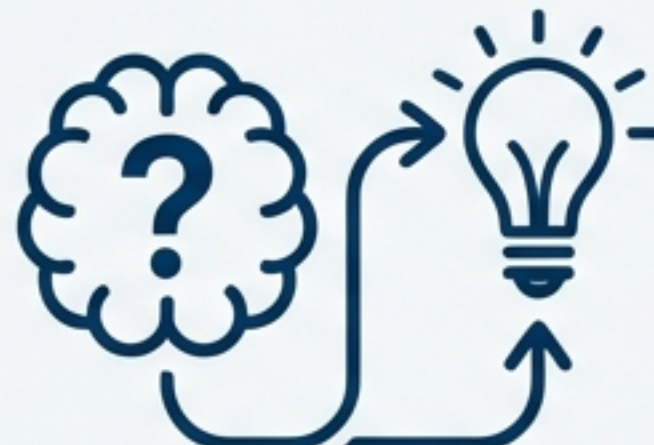
From 'What' to 'Why' to 'How': A Three-Part Investigation

This project moves beyond a standard analysis. We document the complete journey from initial data exploration to rigorous causal reasoning and finally to a production-ready machine learning solution.



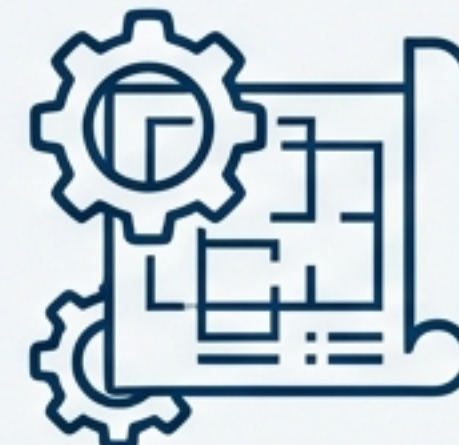
Exploratory Data Analysis (The 'What')

Uncovering the structure, trends, and hidden puzzles within a massive dataset of 22,177 images and 54,031 aircraft annotations.



Causal Inference (The 'Why')

Answering a critical question:
Does an aircraft's size in an image
causally affect its perceived
rarity, or is it merely correlation?



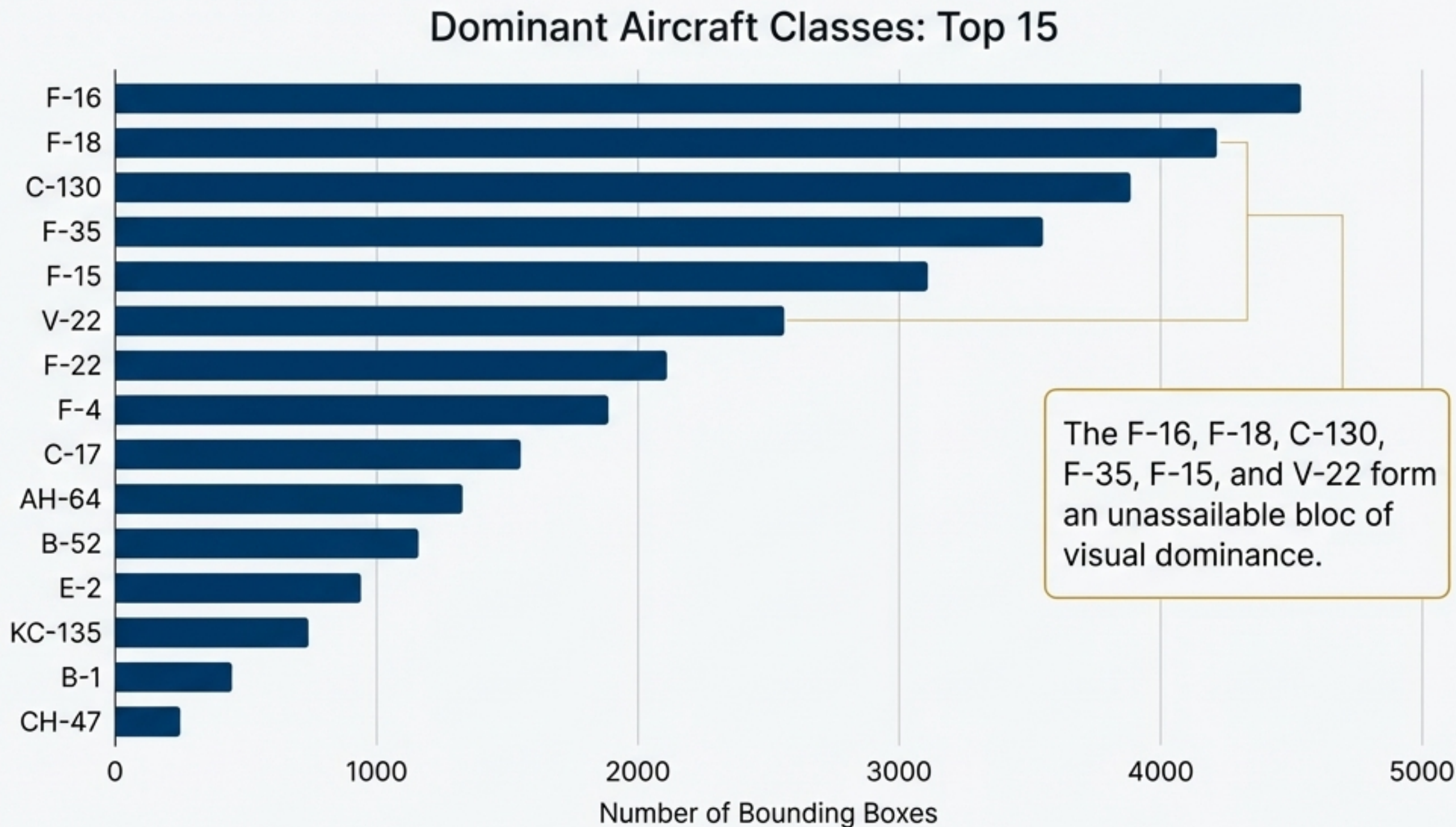
Machine Learning (The 'How')

Building and deploying a high-performance YOLOv8 classifier to put our insights into practice.

A Tale of Two Air Forces: A Few Common Jets Dominate a Long Tail of Rare Assets

Just 15 of the 96 aircraft types monopolize a staggering **41.2%** of all bounding box annotations.

This isn't a flaw; it's a data-driven portrait of global airpower concentration and operational rarity.



The Size-Rarity Puzzle: A Weak Correlation Uncovers a Deeper Question

Initial Observation

Our exploration revealed a curious pattern: larger bounding boxes appeared more frequently in the rare aircraft classes.

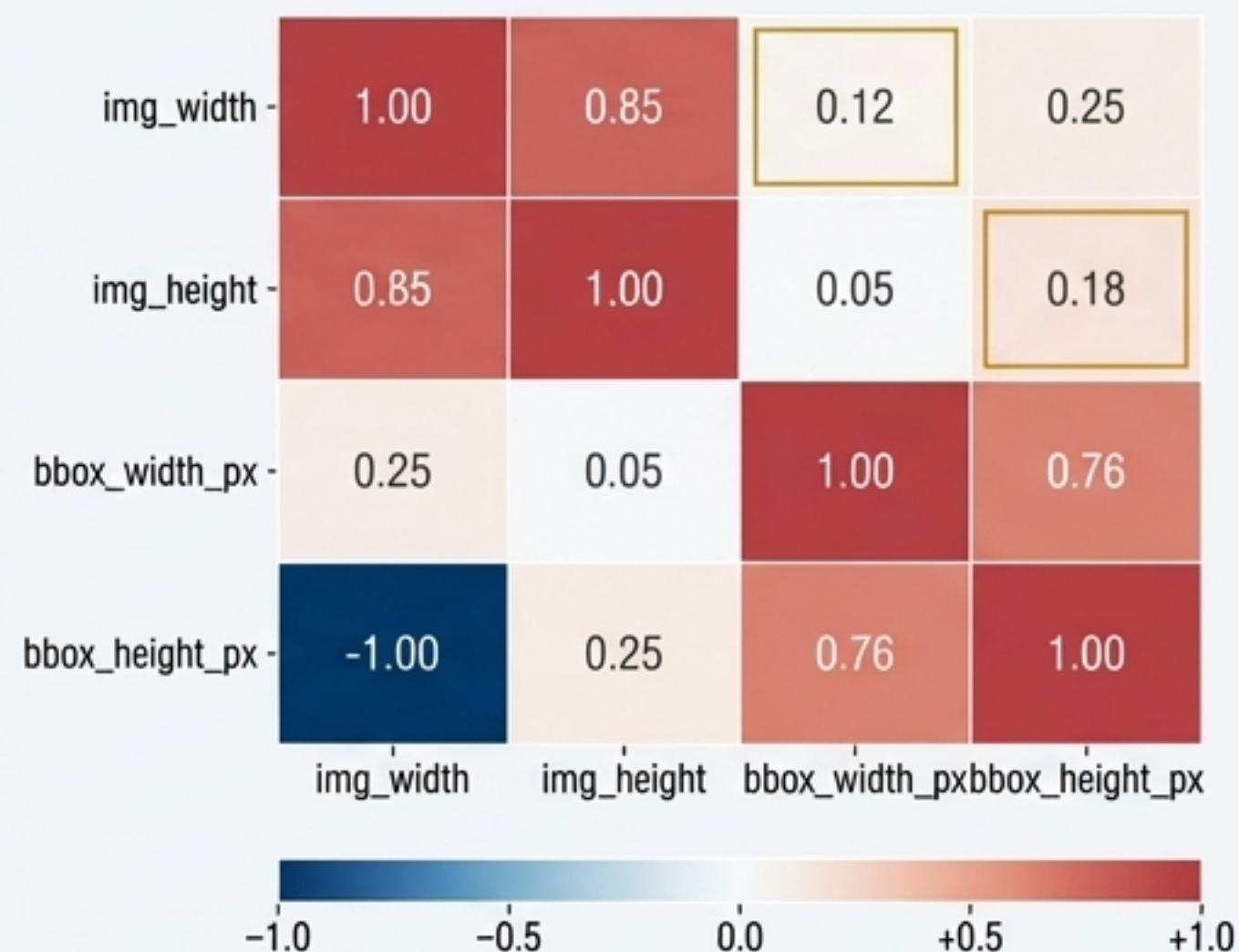
The Obvious Hypothesis

“Larger bounding boxes simply come from higher-resolution images.”

The Data's Verdict

This hypothesis is incorrect. The correlation between image resolution and bounding box size is surprisingly weak.

Correlation Matrix: Image & Bounding Box Dimensions



Bounding box size reflects genuine variation in aircraft scale and photography context, not just image resolution. This validates it as a meaningful variable for causal analysis.

The Central Question: Is the Link Between Size and Rarity Causal?

The pattern we found forces a critical question: **Does having a large bounding box *cause* an aircraft to be from a rare class, or is this association driven by hidden confounding factors?**

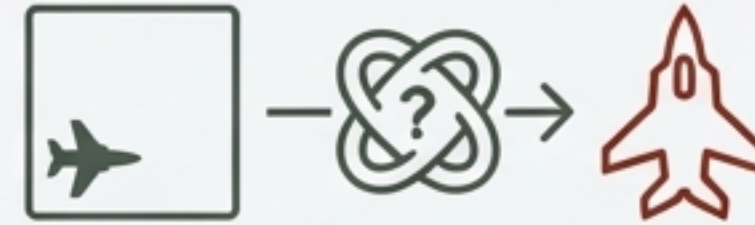
Why This Matters

If the link is **CAUSAL**



Our modeling strategy should directly address this. For example, we should oversample small-scale images of rare aircraft during training to break the learned dependency.

If the link is **SPURIOUS**



The association is a distraction. It's due to other factors (like specific camera types used for rare jets), and we should not base our modeling strategy on it.

Traditional regression cannot answer this. We must use the tools of causal inference.

Isolating Causality with Propensity Score Matching (PSM)

Core Idea

PSM is a technique from econometrics that creates fair, apples-to-apples comparisons. It allows us to simulate a randomized controlled trial using observational data.

The Confounding Problem



"Observing that ice cream sales and drowning deaths are correlated doesn't mean ice cream causes drowning. The summer heat is a **confounder** that drives both. PSM helps us control for 'the weather'."

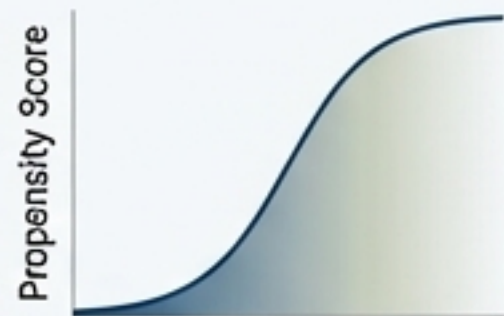
1. Define Groups



Confounded Groups

Start with two confounded groups (e.g., images with small vs. large aircraft).

2. Estimate Propensity



Confounders (e.g., Resolution)

Train a model to predict the probability (Propensity Score) of having a large bounding box, based only on confounders like image resolution and aspect ratio.

3. Match Pairs



For each aircraft with a large bounding box, find one with a small box that has a nearly identical Propensity Score.

4. Compare Outcomes



Balanced Groups

The result is two new groups that are statistically balanced on all confounders. Any remaining difference in outcomes can be attributed to the treatment itself.

The Causal Link is Real: Large Bounding Boxes Increase Perceived Rarity

The Naïve Difference (Confounded)

+7.59

percentage points

Before controlling for confounders, aircraft in large bounding boxes are 7.59 percentage points more likely to be from a rare class.

The Causal Effect (After Matching)

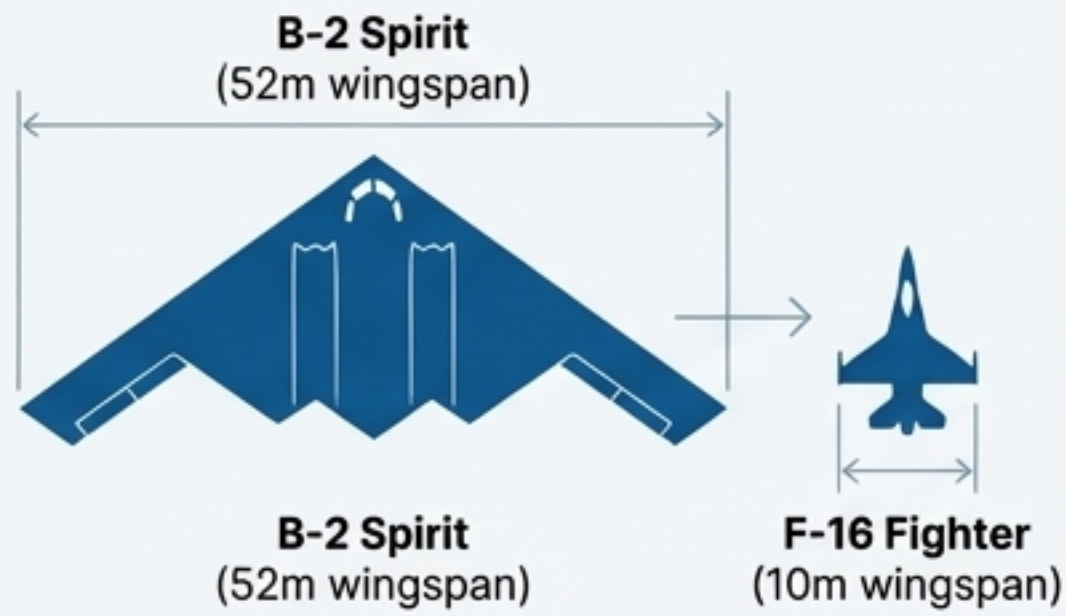
+8.30

percentage points

After creating a balanced comparison with PSM, the true causal effect is revealed. Large bounding boxes causally increase the probability of an aircraft being from a rare class by 8.30 percentage points.

The Causal Pathway: Why Size is Not Accidental

The +8.30pp effect is not a statistical artifact. It is driven by three distinct real-world mechanisms that link aircraft type, operational context, and how they are photographed.



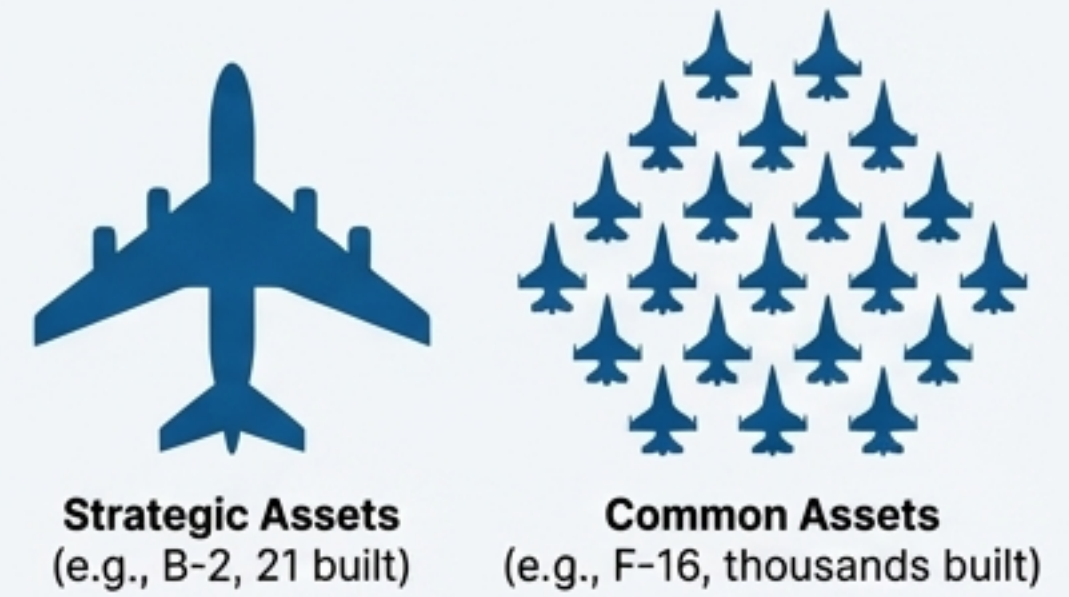
1. Physical Size

Rare strategic assets are physically massive. A B-2 Spirit bomber (52m wingspan) dwarfs a common F-16 fighter (10m wingspan).



2. Photography Intent

Rare, high-value assets are photographed deliberately and at close range for intelligence purposes. Common aircraft are often captured incidentally during routine surveillance.



3. Operational Rarity

Strategic assets like the B-2 (21 built) are deployed far less frequently than workhorses like the F-16 (thousands built). When they do appear, they are high-priority targets.

From Causal Insight to a Production-Ready Classifier

The Goal

To build a practical model that can accurately classify aircraft, informed by our understanding of the dataset's complexities.

Why YOLOv8-cls?

While initial tests with ResNet-50 faced memory and convergence issues, YOLOv8's classification variant offered a superior solution.

Key Advantages of YOLOv8-cls

- ✓ **Optimized for Speed:** Designed for real-time performance.
- ✓ **Robust to Imbalance:** Handles long-tail distributions effectively.
- ✓ **Lightweight:** 1.44M parameters vs. 25M for ResNet-50.

Setup



Focus: Top 6 most frequent classes for rapid, meaningful iteration.



Dataset: 9,240 clean, cropped aircraft images.



Split: Stratified 70% Train / 15% Validation / 15% Test.

High-Accuracy, High-Speed Performance on Unseen Data

10 Epochs completed in **5.3 minutes** on a single Tesla T4 GPU.

Top-1 Accuracy

85.4%

The model's single best prediction is correct on the unseen test set.
(Random chance would be 16.7%)

Top-5 Accuracy

99.6%

The correct aircraft class appears in the model's top 5 predictions for 99.6% of test images.

Test accuracy (**85.4%**) being slightly higher than validation accuracy (84.5%) indicates strong generalization and no overfitting.



0.8ms per image
(~1,250 images per second)

The Model in Action: High-Confidence Predictions



True: **F-35** | Predicted: **F-35**
(**99.9%** confidence)

✓ **CORRECT**



True: **C-130** | Predicted: **C-130**
(**98.7%** confidence)

✓ **CORRECT**



True: **V-22** | Predicted: **V-22**
(**100.0%** confidence)

✓ **CORRECT**

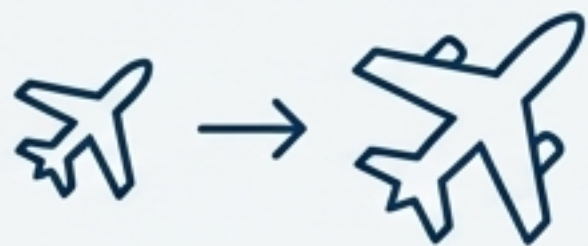
The model demonstrates extremely high confidence in its correct predictions, suggesting it has learned powerful, discriminative features for each class.

The Future of AI is Causal

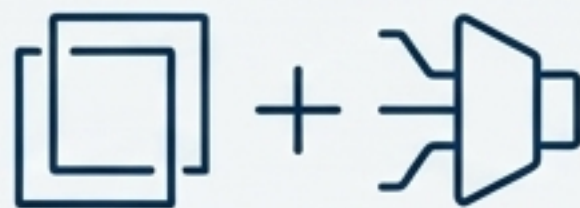
“Large bounding boxes don’t just happen to occur with rare aircraft—they are causally linked through the intrinsic properties of strategic military assets (their physical size, photography intent, and operational rarity).”

Machine learning is often seen as a black box. This project proves that by asking “Why?” and using causal inference, we can move beyond pattern recognition to scientific understanding. This understanding leads directly to building smarter, more robust, and more reliable systems.

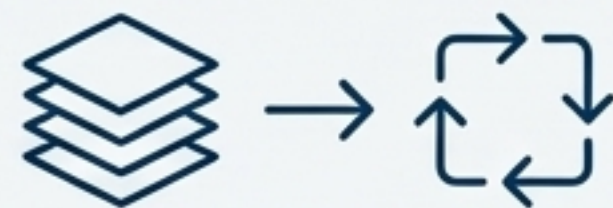
How Causality Improves ML



Design better data augmentation (e.g., artificially breaking the size-rarity link).



Engineer more informative features (e.g., using bbox size as a direct input).



Build superior training strategies (e.g., stratified sampling by size **and** class).